Bond & Stock Valuation

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Key Phrase

Value any asset by discounting its expected future cash flows back at an appropriate, risk adjusted discounted rate

$$Price = PV(Future\ Cash\ Flows)$$
$$= \sum_{t=0}^{\infty} \frac{CF_t}{(1+r)^t}$$

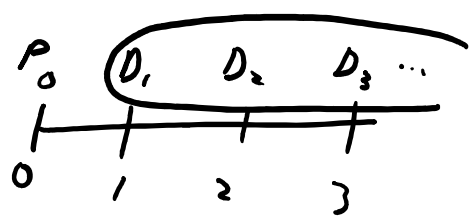
- Two ingredients
  - Cash flows
  - Discount rate

Bond Valuation Review

- Cash flows
  - Coupons
  - Face value
- Discount rate: yield to maturity

$$
\begin{aligned}
Price &= PV(Coupons) + PV(Face) \\
&= Coupon \times ADF(n, r) + \frac{Face}{(1+r)^n} \\
&= PV(r, n, PMT = Coupon, FV = Face)
\end{aligned}
$$

Stock Valuation Review



- P_0 stock price today
- D_1 dividend at time 1
- r discount rate (i.e. cost of equity)
- Assume that dividends will grow at a constant rate, g, forever
  - $D_{t+1} = D_t \times (1+g)$
- Since future cash flows grow at a constant rate forever, the value of a constant growth stock is the present value of a growing perpetuity

$$P_0 = \frac{D_1}{r-g}$$

Bond vs Stock Valuation

TVM	Bond	Stock
Time Frame n	Time to Maturity	Investment horizon
Discount Rate r	YTM / IRR	Cost of Equity
Present Value	Bond Price today	Stock Price today
Future Value	Face Value of Bond	Perpetuity (Terminal) Value
Payments	Coupon Payments	Dividends

Bond vs Stock Features

Debt

- Not an ownership interest.
- Creditors do not have voting rights.
- Interest is considered a cost of doing business and is tax deductible (up to 30% of EBIT).
- Creditors have legal recourse if interest or principal payments are missed.
- Excess debt can lead to financial distress and bankruptcy.

Equity

- Ownership interest.
- Common stockholders vote for the board of directors and other issues.
- Dividends are not considered a cost of doing business and are not tax deductible.
- Dividends are not a liability of the firm, and stockholders have no legal recourse if dividends are not paid.
- An all-equity firm cannot go bankrupt.

Bond Terminology

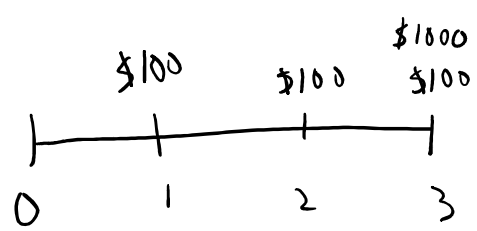
- Maturity date
  - Date the final payment is to be received
- Par (or face) value
  - The amount that will be received at maturity
- Coupon payments
  - The regular cash flow that the bond issuer pays
  - Coupon = Coupon Rate × Face
- Yield to Maturity
  - The market return of a bond given the bond's price, coupon payments, par value and timing of payments

Cash Flows

Discount Rate

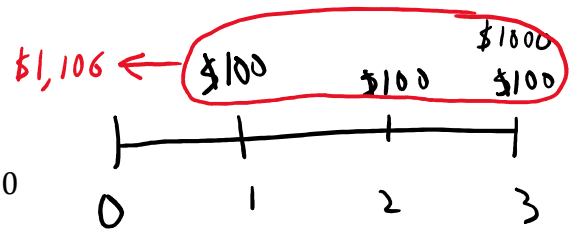
Bond Valuation Example

- A bond has a face value of \$1000. The bond pays annual coupons at a coupon rate of 10%. The bond has a maturity of 3 years and it's yield to maturity is 6%. What is the price of the bond?
- Discount rate: ytm=6%
- Cash flows:
  - Annual coupon: \$1000 × 10% = \$100
  - Face value payment \$1000 @ year 3



Bond Valuation Example

- Discount rate: ytm=6%
- Cash flows:
  - Annual coupon: $\$1000 \times 10\% = \$100$
  - Face value payment $1000 @ year 3



$$
\begin{aligned}
Price &= PV(Coupons) + PV(Face) \\
&= \$100 \cdot ADF(3{,}6\%) + \frac{\$1000}{(1+6\%)^3} \\
&= PV(r = 6\%, n = 3, PMT = 100, FV = 1000) \\
&= \$1{,}106.92
\end{aligned}
$$

Bond Valuation

- Cash flows
  - Coupons
  - Face value
- Discount rate: yield to maturity

$$
\begin{aligned}
Price &= PV(Coupons) + PV(Face) \\
&= Coupon \times ADF(n, r) + \frac{Face}{(1+r)^n} \\
&= PV(r, n, PMT = Coupon, FV = Face)
\end{aligned}
$$

YTM & IRR

A bond has a face value of \$1000. It has a 10% coupon rate and pays annual coupons. The bond has a maturity of 3 years and it's yield to maturity is 6%. The price of the bond is \$1,106.92.

$$\$1{,}106.92 = \frac{\$100}{(1+6\%)^1} + \frac{\$100}{(1+6\%)^2} + \frac{\$100}{(1+6\%)^3} + \frac{\$1000}{(1+6\%)^3}$$

- YTM and coupon rates are quoted as APRs
  - Annual coupons
  - YTM = 6%
- IRR is the per-period discount rate
  - IRR = 6%

YTM & IRR

A bond has a face value of \$1000. It has a 10% coupon rate and pays semiannual coupons. The bond has a maturity of 3 years and it's yield to maturity is 6%. The price of the bond is \$1,108.34.

- Semiannual coupon rate: $5\% = \frac{10\%}{2}$
- Semiannual coupon: $\$50 = 5\% \times \$1000$
- YTM is quoted as an APR: YTM = 6%
- IRR is the per-period discount rate: $3\% = \frac{6\%}{2}$

$$\$1{,}108.34 = \frac{\$50}{(1+3\%)^1} + \frac{\$50}{(1+3\%)^2} + \cdots + \frac{\$50}{(1+3\%)^6} + \frac{\$1000}{(1+3\%)^6}$$

Yield Curve

Spot rates are the yields of zero coupon bonds

- Reflect the time value of money for a single time period
- Also called "yields"

Let us assume that we have 3 zero-coupon bonds, each with a par value of $1,000, and info as follows:

- 1-year bond with a market price of $925.93
- 2-year bond with a market price of $841.75
- 3-year bond with a market price of $758.33

From these bond prices, we can calculate the yield to maturity for each bond

$$Price = \frac{Par}{(1 + yield)^n}$$

Yield Curve

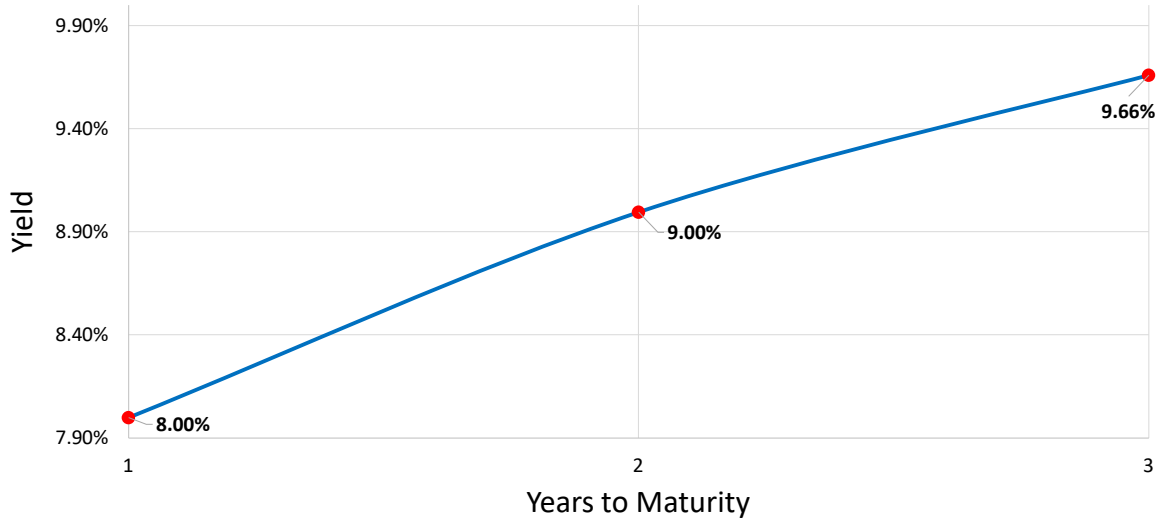
We can now calculate the yield to maturity as follows:

- For the 1-year bond: $\$925.93 = \frac{\$1000}{(1+y_1)^1} \Rightarrow y_1 = 8\%$
- For the 2-year bond: $\$841.75 = \frac{\$1000}{(1+y_2)^2} \Rightarrow y_2 = 9\%$
- For the 3-year bond: $\$758.33 = \frac{\$1000}{(1+y_3)^3} \Rightarrow y_3 = 9.66\%$

Yield Curve

- Plot spot rates against maturity

Yield Curve



Interest Rate Risk

- Interest rate risk measures how sensitive a bond's price is to changes interest rates
- Bonds with longer maturities and lower coupon rates have higher interest rate risk
  - Intuition: Cash flows further in the future are riskier
  - Mathematically: Small changes in interest rates compound into large effects in the future

Problem 1

Deacon has a corporate bond outstanding that has 12 years remaining to maturity, semiannual coupon payments, a coupon rate of 12% per year and a yield-to-maturity of 7.65% per year. The bond has a $1000 face value. What is the price of the bond?

$$P_0 = \$60 \times ADF\left(12 \times 2, \frac{7.65\%}{2}\right) + \frac{\$1{,}000}{\left(1 + \frac{7.65\%}{2}\right)^{12\times 2}}$$

$$= \$1337.64$$

Excel

- $PV\left(r = \frac{7.65\%}{2}, n = 12 \times 2, PMT = \$60, FV = \$1{,}000\right) = \$1337.64$

Problem 2

A bond has 5 years to maturity. It has a 5.5% coupon rate and pays semi-annual coupon payments. Its face value is $1,000 and has a market price of $1,032.19. Assume that the next coupon payment is exactly six months away.

a) What is the yield-to-maturity of the bond?

$$Coupon = \frac{5.5\%}{2} \times \$1{,}000$$

$$= \$27.50$$

$$\$1{,}032.19 = \$27.50 \times \left[\frac{1 - \frac{1}{(1 + IRR)^{5\times 2}}}{IRR}\right] + \frac{\$1{,}000}{(1 + IRR)^{5\times 2}}$$

$$\Rightarrow IRR = 2.384\%$$

Excel

- $RATE(n = 5 \times 2, PMT = \$27.50, PV = -\$1{,}032.19, FV = \$1{,}000) = 2.384\%$

Problem 2

A bond has 5 years to maturity. It has a 5.5% coupon rate and pays semi-annual coupon payments. Its face value is $1,000 and has a market price of $1,032.19. Assume that the next coupon payment is exactly six months away.

a) What is the yield-to-maturity of the bond?

$$IRR = 2.384\%$$

$$\begin{aligned} YTM &= IRR \times 2 \\ &= 2.384\% \times 2 \\ &= 4.769\% \end{aligned}$$

Problem 2

A bond has 5 years to maturity. It has a 5.5% coupon rate and pays semi-annual coupon payments. Its face value is $1,000 and has a market price of $1,032.19. Assume that the next coupon payment is exactly six months away.

b) What is the effective annual rate implied by this price?

$$EAR = (1 + 2.384\%)^2 - 1 = 4.826\%$$

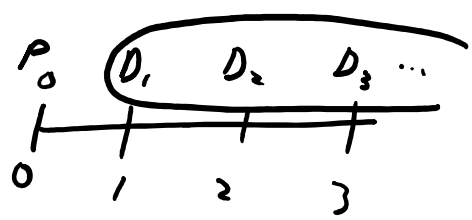
Stock Valuation

- Cash flows: dividends
- Discount rate: r_E cost of equity

$$Price = PV(Future\ Dividends)$$
$$= \sum_{t=0}^{\infty} \frac{D_t}{(1+r_E)^t}$$

- Valuation of different types of stocks
  - Constant Growth
  - Differential Growth

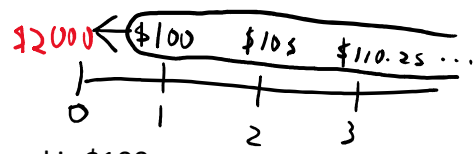
Constant Growth



- P_0 stock price today
- D_1 dividend at time 1
- r discount rate (i.e. cost of equity)
- Assume that dividends will grow at a constant rate, g, forever
  - $D_{t+1} = D_t \times (1+g)$
- Since future cash flows grow at a constant rate forever, the value of a constant growth stock is the present value of a growing perpetuity

$$P_0 = \frac{D_1}{r-g}$$

Constant Growth



- Example
  - Receive dividends forever, first dividend is $100
  - Dividends grow at 5%
  - Discount rate 10%

$$
\begin{aligned}
P_0 &= \frac{D_1}{(1+r)^1} + \frac{D_2}{(1+r)^2} + \cdots \\
&= \frac{\$100}{(1+10\%)^1} + \frac{\$105}{(1+10\%)^2} + \cdots \\
&= \frac{D_1}{r-g} \\
&= \frac{\$100}{10\% - 5\%} \\
&= \$2000
\end{aligned}
$$

Generic TVM

Perpetuity w/ growth

Sources of Return

- Stock ownership produces cash flows from
  - Dividends
  - Capital Gains

- The discount rate can be broken into two parts
  - The dividend yield $\frac{D_{t+1}}{P_t}$
  - Capital gains yield (price appreciation) g

$$P_0 = \frac{D_1}{r-g} \Leftrightarrow r = \frac{D_1}{P_0} + g$$

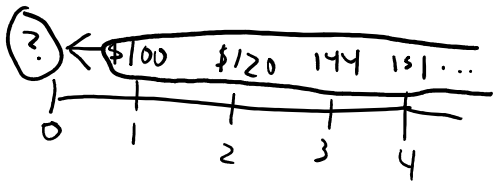
Differential Growth

- Assume that dividends will grow at different rates in the short run and then will grow at a constant rate thereafter
- To value a stock, we need to:
  1. Estimate the future short run dividends
  2. Estimate the future stock price when the stock becomes a constant growth stock
  3. Compute the total present value of the estimated future dividends and future stock price at the appropriate discount rate

Differential Growth

- Example
  - Receive dividends forever, first dividend is $100
  - Dividends grow at 20% for years 2 and 3
  - Dividends grow at 5% thereafter
  - Discount rate 10%

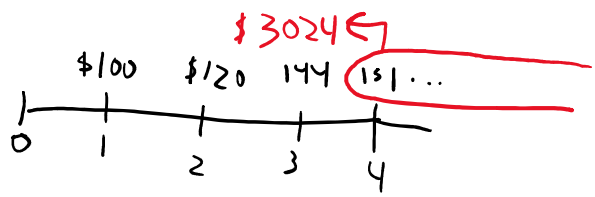
$$D_2 = D_1 \times (1 + g) = \$100 \times (1 + 20\%) = \$120$$
$$D_3 = D_2 \times (1 + g) = \$120 \times (1 + 20\%) = \$144$$
$$D_4 = D_3 \times (1 + g) = \$144 \times (1 + 5\%) = \$151$$



Differential Growth

- Long-run perpetuity value

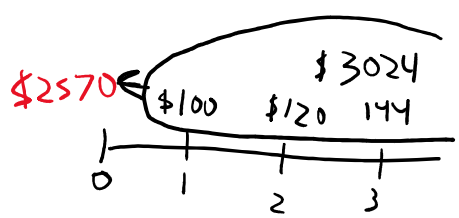
$$
\begin{aligned}
P_3 &= \frac{D_4}{r-g} \\
&= \frac{\$151}{10\% - 5\%} \\
&= \$3024
\end{aligned}
$$



Differential Growth

- Total stock value today

$$
\begin{aligned}
P_0 &= \frac{D_1}{(1+r)^1} + \frac{D_2}{(1+r)^2} + \frac{D_3}{(1+r)^3} + \frac{P_3}{(1+r)^3} \\
&= \frac{\$100}{(1+10\%)^1} + \frac{\$120}{(1+10\%)^2} + \frac{\$144}{(1+10\%)^3} + \frac{\$3024}{(1+10\%)^3} \\
&= \$2570
\end{aligned}
$$



Problem 3

Deacon currently pays no dividends and instead reinvests all earnings in the firm. In year 6, they will begin paying a dividend of $5.00 per share. Dividends will grow at 2% per year in perpetuity following that. The required return is 15% per year.

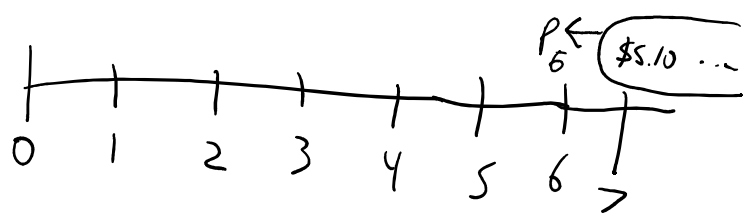
- a) What is the ex-dividend stock price at year 6? Cum-dividend price at year 6?
- b) What is the stock price at year 5?
- c) What is the stock price today?

Problem 3

- a) What is the ex-dividend stock price at year 6?

$$P_6 = \frac{D_7}{r - g}$$

$$\begin{aligned} D_7 &= D_6 \times (1 + g) \\ &= \$5 \times (1 + 2\%) \\ &= \$5.10 \end{aligned}$$



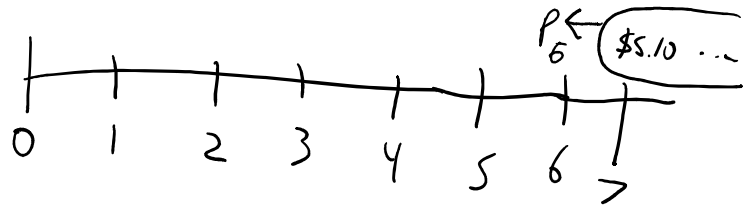
$$\begin{aligned} P_6 &= \frac{\$5.10}{15\% - 2\%} \\ &= \$39.23 \end{aligned}$$

Problem 3

- a) What is the cum-dividend stock price at year 6?

$$P_6^{Cum} = P_6^{Ex} + D_6$$

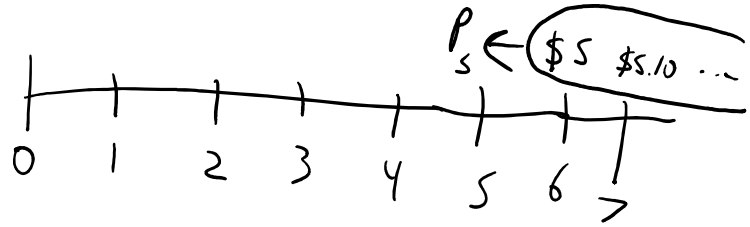
$$= \$39.23 + \$5$$
$$= \$44.23$$



Problem 3

- b) What is the stock price at year 5?

$$P_5 = \frac{\$5}{15\% - 2\%}$$
$$= \$38.46$$

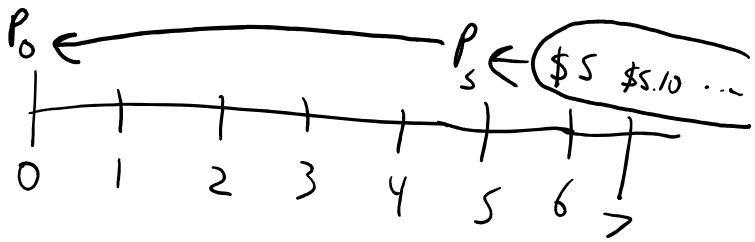


Problem 3

- c) What is the stock price today?

$$P_5 = \frac{\$5}{15\% - 2\%}$$
$$= \$38.46$$

$$P_0 = \frac{\$38.46}{(1 + 15\%)^5}$$
$$= \$19.12$$



Problem 4

Wake, Inc. just paid a dividend of $2.50 per share. The dividends are expected to grow for the next 3 years at 8% per year, then grow at 3% per year forever. The cost of equity for Wake stock is 12% per year.

a) What is Wake's stock price?

b) What is the ex-dividend stock price in year 2?

c) If you purchased the share of Wake at time 2, at the price you calculated in (b), and held it for one year, what should the return on investing in Wake stock be? What portion of that return comes from dividends, and what portion from capital gain?

Problem 4

Wake, Inc. just paid a dividend of \$2.50 per share. The dividends are expected to grow for the next 3 years at 8% per year, then grow at 3% per year forever. The cost of equity for Wake stock is 12% per year.

a) What is Wake's stock price?

$$P_0 = \frac{D_1}{(1+r)^1} + \frac{D_2}{(1+r)^2} + \frac{D_3}{(1+r)^3} + \frac{D_4}{(1+r)^4} + \cdots$$

Problem 4

Dividends

$$\begin{aligned}
D_1 &= D_0 \times (1+g) = \$2.50 \times (1+8\%) = \$2.70 \\
D_2 &= D_1 \times (1+g) = \$2.70 \times (1+8\%) = \$2.92 \\
D_3 &= D_2 \times (1+g) = \$2.92 \times (1+8\%) = \$3.15 \\
D_4 &= D_3 \times (1+g) = \$3.15 \times (1+3\%) = \$3.24
\end{aligned}$$

Perpetuity

$$\begin{aligned}
P_3 &= \frac{D_4}{r-g} \\
&= \frac{\$3.24}{12\% - 3\%} \\
&= \$36.04
\end{aligned}$$

Problem 4

$$
\begin{aligned}
P_0 &= \frac{D_1}{(1+r)^1} + \frac{D_2}{(1+r)^2} + \frac{D_3}{(1+r)^3} + \frac{P_3}{(1+r)^3} + \cdots \\
&= \frac{\$2.70}{(1+12\%)^1} + \frac{\$2.92}{(1+12\%)^2} + \frac{\$3.15}{(1+12\%)^3} + \frac{\$36.04}{(1+12\%)^3} \\
&= \$32.63
\end{aligned}
$$

Problem 4

Wake, Inc. just paid a dividend of \$2.50 per share. The dividends are expected to grow for the next 3 years at 8% per year, then grow at 3% per year forever. The cost of equity for Wake stock is 12% per year.

b) What is the ex-dividend stock price in year 2?

$$
\begin{aligned}
P_2 &= \frac{D_3}{(1+r)^1} + \frac{D_4}{(1+r)^2} + \cdots \\
&= \frac{D_3}{(1+r)^1} + \frac{P_3}{(1+r)^1} \\
&= \frac{\$3.15}{(1+12\%)^1} + \frac{\$36.04}{(1+12\%)^1} \\
&= \$34.99
\end{aligned}
$$

Problem 4

c) If you purchased the share of Wake at time 2, at the price you calculated in (b), and held it for one year, what should the return on investing in Wake stock be? What portion of that return comes from dividends, and what portion from capital gain?

$$\begin{aligned}
\text{Return} &= \text{Capital Gains} + \text{Dividend Yield} \\
&= \frac{P_3 - P_2}{P_2} + \frac{D_3}{P_2} \\
&= \frac{\$36.04 - \$34.99}{\$34.99} + \frac{\$3.15}{\$34.99} \\
&= 3\% + 9\% \\
&= 12\%
\end{aligned}$$